

TUNKA, Russian Federation

WMO: 308110

Lat: 51.733N

Lon: 102.533E

Elev: 2365

StdP: 13.48

Time Zone: 8.00 (E08)

Period: 82-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-30.0	-37.5	0.7	-33.6	-33.6	0.9	-29.7	15.5	2.0	13.1	6.3	0.0	330	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	80.9	62.6	77.6	61.3	74.4	59.9	65.1	76.4	63.3	73.9	61.7	71.4	4.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	88.1	68.5	59.3	82.3	67.1	57.8	77.8	65.7	31.5	76.6	30.1	74.6	28.8	72.0	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	13.8	11.5	-38.1	86.7	4.5	3.6	-41.4	89.3	-44.0	91.5	-46.5	93.5	-49.8	96.1		
(5)			-38.4	67.4	4.3	2.1	-41.5	68.9	-44.1	70.2	-46.5	71.4	-49.7	72.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.0	-12.1	-4.5	16.5	35.7	48.7	57.9	62.6	59.5	47.3	32.5	9.6	-7.7		
	DBStd	27.70	9.09	10.01	11.74	7.47	7.24	6.03	4.70	5.01	6.39	7.98	11.35	9.97		
	HDD50	8716	1925	1526	1038	432	114	7	1	3	126	543	1211	1790		
	HDD65	13184	2390	1946	1503	880	507	222	101	179	532	1008	1661	2255		
	CDD50	1058	0	0	0	2	74	245	393	299	44	1	0	0		
	CDD65	48	0	0	0	0	1	10	27	10	0	0	0	0		
	CDH74	755	0	0	0	0	84	197	325	146	3	0	0	0		
	CDH80	134	0	0	0	0	14	39	66	15	0	0	0	0		
	WSAvg	2.7	0.8	1.3	2.8	4.9	4.7	4.2	3.1	2.9	3.1	2.4	1.6	0.9		
	PrecAvg	14.6	0.2	0.1	0.2	0.5	1.0	2.3	4.0	3.4	1.9	0.5	0.3	0.2		
(15) Precipitation	PrecMax	20.3	0.4	0.5	0.4	1.5	2.3	4.3	7.4	5.9	3.9	1.7	0.8	0.6		
	PrecMin	9.8	0.0	0.0	0.0	0.1	0.2	0.8	1.0	1.1	0.6	0.0	0.0	0.0		
	PrecStd	2.4	0.1	0.1	0.1	0.4	0.5	0.9	1.4	1.1	0.8	0.4	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	31.1	53.2	68.3	82.2	84.8	85.9	82.3	72.5	60.7	39.8	23.4		
		MCWB	17.7	26.7	40.5	49.8	59.0	63.2	66.1	64.5	56.9	46.8	33.2	21.0		
	2%	DB	12.2	21.8	44.4	60.6	76.1	79.3	81.3	77.9	67.4	54.5	34.3	16.9		
		MCWB	10.9	19.0	35.3	45.6	55.8	60.2	64.1	63.0	53.9	43.3	30.2	15.2		
	5%	DB	7.1	17.3	39.1	55.8	70.8	75.5	77.6	74.3	63.4	49.5	29.7	11.8		
		MCWB	6.0	15.0	31.9	42.1	52.6	59.1	63.1	61.4	52.2	40.8	26.8	10.7		
	10%	DB	2.6	12.8	34.5	50.7	65.8	71.7	74.2	71.0	59.4	45.1	26.1	7.2		
		MCWB	1.7	11.0	29.1	39.4	50.4	57.2	61.5	59.6	50.3	38.3	23.6	6.2		
	0.4%	WB	18.5	26.4	41.1	50.5	60.6	65.2	68.2	67.0	58.8	47.6	33.8	20.8		
		MCDB	20.5	30.8	52.6	65.5	79.1	79.7	80.8	78.1	68.3	59.3	39.0	22.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	10.8	19.1	35.6	45.8	56.7	62.7	66.0	64.0	55.6	43.9	30.4	15.3		
		MCDB	12.1	21.8	43.5	60.0	74.1	75.0	77.6	74.9	65.2	53.3	34.2	16.8		
	5%	WB	6.1	15.1	32.4	42.7	53.7	60.6	64.3	62.3	53.2	41.2	27.2	10.7		
		MCDB	7.2	17.3	38.6	54.7	68.6	72.3	74.5	72.3	61.7	48.6	29.7	11.8		
	10%	WB	1.8	11.1	29.2	39.7	51.0	58.6	62.6	60.7	51.0	38.6	23.7	6.3		
		MCDB	2.6	12.8	34.5	49.9	64.2	69.5	72.2	69.0	58.5	44.5	25.9	7.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.9	26.4	26.1	22.9	25.3	21.3	19.9	20.0	22.1	20.9	19.5	17.6		
		MCDBR	23.2	29.4	28.5	34.0	37.2	31.1	28.1	27.7	29.8	29.5	21.5	21.8		
		MCWBR	22.0	26.3	19.2	19.7	19.0	14.1	12.5	13.0	16.0	17.9	17.4	20.5		
	5% WB	MCDBR	23.2	29.4	27.7	33.0	33.7	27.1	24.1	24.3	26.2	27.5	21.4	22.1		
		MCWBR	21.9	26.1	18.8	19.3	17.6	12.8	12.2	11.9	14.4	17.0	17.3	20.7		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.276	0.319	0.364	0.408	0.402	0.399	0.368	0.321	0.300	0.274	0.251		
	taud		2.223	2.193	2.167	2.229	2.168	2.310	2.362	2.425	2.499	2.454	2.292	2.183		
	Ebn at Noon		244	268	276	276	268	271	269	273	276	260	231	218		
	Edh at Noon		21	29	37	40	44	39	36	32	27	23	20	18		
(44) All-Sky Solar Radiation	RadAvg		429	770	1220	1561	1712	1749	1624	1416	1125	735	440	310		
	RadStd		17	30	57	74	81	107	125	89	95	41	25	16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page